

VILLARRICA, Paraguay

WMO: 862330

Lat: 25.767S

Lon: 56.433W

Elev: 161

StdP: 99.41

Time Zone: -4.00 (PRY)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	7.2	1.4	4.3	9.8	3.0	4.8	9.9	8.5	20.8	8.1	21.1	1.1	180	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.3	35.5	24.2	34.6	24.3	33.7	24.3	26.7	32.1	26.3	31.8	25.7	31.1	3.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.8	29.8	24.7	20.1	29.3	24.3	19.6	28.8	84.9	31.6	83.2	32.7	80.8	30.7	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.9	5.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	27.5	26.7	26.1	23.2	19.1	18.9	18.3	19.4	20.6	23.5	24.3	26.6
	DBStd	5.05	1.78	2.40	2.48	3.47	3.91	4.78	5.19	5.69	4.86	3.42	2.99	2.38
	HDD10.0	7	0	0	0	0	1	1	3	3	0	0	0	0
	HDD18.3	260	0	0	0	4	38	54	69	60	31	3	1	0
	CDD10.0	4694	542	468	500	397	284	268	262	293	319	418	429	513
	CDD18.3	1904	283	235	242	152	63	71	69	92	100	163	180	255
	CDH23.3	17659	3072	2385	2346	1178	337	364	419	774	925	1476	1696	2688
	CDH26.7	7201	1387	1043	1025	413	71	78	104	292	361	587	668	1171
Wind	WSAvg	1.9	1.4	1.5	1.3	1.8	2.1	2.1	2.3	2.5	2.2	2.3	1.9	1.5
Precipitation	PrecAvg	1650	172	137	147	163	152	106	75	86	107	182	169	150
	PrecMax	2574	402	449	344	453	434	361	235	284	357	431	505	497
	PrecMin	1072	6	32	24	0	23	3	1	7	0	3	22	29
	PrecStd	350	89	87	71	101	100	74	60	69	74	85	101	92
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.2	36.8	36.1	33.8	30.9	30.0	30.8	33.4	35.4	35.4	35.1	36.2
		MCWB	23.7	24.4	23.4	24.5	23.1	21.9	20.6	20.7	22.2	25.1	25.2	24.6
	2%	DB	34.9	34.8	34.4	32.6	29.1	28.7	29.3	31.9	33.3	33.7	33.9	34.7
		MCWB	24.0	24.8	24.3	24.5	22.1	21.7	20.1	20.6	22.2	24.3	24.5	24.9
	5%	DB	33.9	33.5	33.1	31.2	27.3	27.4	27.8	30.3	30.9	32.0	32.4	33.5
		MCWB	24.2	24.9	24.3	24.2	21.4	21.3	19.8	20.4	21.3	23.7	23.9	25.0
	10%	DB	32.8	32.1	31.8	29.4	25.5	25.5	25.8	28.1	28.5	30.1	30.7	32.2
		MCWB	24.5	24.8	23.9	23.5	20.5	20.7	19.6	19.7	20.2	22.7	23.3	24.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	27.4	26.7	26.5	24.4	23.7	23.0	23.7	24.8	26.4	26.7	27.5
		MCDB	32.4	32.3	32.2	32.1	28.6	27.6	27.7	29.7	31.8	33.4	32.4	32.8
	2%	WB	26.4	26.5	26.0	25.4	23.3	22.6	21.8	22.3	23.5	25.3	25.9	26.6
		MCDB	31.7	31.6	31.4	30.7	27.5	26.8	26.2	28.8	30.4	31.9	31.7	32.1
	5%	WB	25.9	25.9	25.4	24.7	22.3	21.7	20.7	21.2	22.2	24.5	25.1	25.9
		MCDB	31.0	31.1	30.6	29.5	25.7	25.9	25.5	27.5	28.8	30.3	30.3	31.4
	10%	WB	25.5	25.4	24.7	24.1	21.2	20.9	19.8	20.2	21.1	23.7	24.3	25.4
		MCDB	30.4	30.2	29.6	28.3	24.5	24.9	24.7	26.1	26.9	28.7	29.1	30.4
Mean Daily Temperature Range	5% DB	MDBR	10.3	9.4	10.1	8.9	8.7	8.5	9.4	10.5	10.7	9.9	10.0	9.8
		MCDBR	11.8	11.4	11.7	10.9	10.1	9.3	10.5	11.7	12.5	11.8	11.9	11.5
		MCWBR	4.0	4.1	4.1	4.5	5.1	4.7	4.8	5.1	5.7	5.2	5.3	4.5
	5% WB	MCDBR	9.6	9.4	9.8	9.6	8.4	8.6	9.8	11.2	11.0	10.2	10.0	9.9
		MCWBR	4.3	4.3	4.4	4.8	5.1	5.0	5.6	6.2	5.7	5.2	5.1	4.7
Clear-Sky Solar Irradiance	taub	0.407	0.410	0.401	0.398	0.374	0.398	0.395	0.498	0.582	0.488	0.417	0.416	
	taud	2.399	2.396	2.404	2.385	2.421	2.347	2.338	2.003	1.799	2.119	2.327	2.358	
	Ebn at Noon	937	919	898	852	828	777	796	740	723	838	922	931	
	Edh at Noon	128	125	118	110	97	100	104	159	213	163	136	133	
All-Sky Solar Radiation	RadAvg	6.65	6.09	5.42	4.37	3.26	2.77	3.19	3.85	4.50	5.30	6.33	6.69	
	RadStd	0.48	0.48	0.38	0.43	0.45	0.26	0.34	0.26	0.48	0.62	0.47	0.50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page